Qiskit_Basic

March 31, 2026

0.1 Differences qasm, statevector, and unitary simulators in qiskit

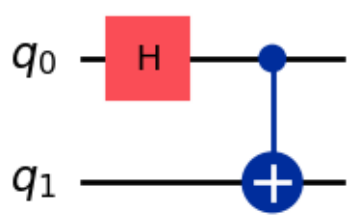
[74]:
```
from qiskit import QuantumCircuit,transpile
from qiskit_aer import Aer, AerSimulator
from qiskit.visualization import array_to_latex
from qiskit.visualization import␣
↪plot_histogram,plot_state_qsphere,plot_bloch_multivector,plot_state_hinton,plot_state_paulive
from qiskit_aer.noise import NoiseModel, depolarizing_error
from math import pi
from qiskit.quantum_info import DensityMatrix, Operator
from qiskit.visualization import plot_state_city
print(Aer.backends())
```

```
[AerSimulator('aer_simulator'), AerSimulator('aer_simulator_statevector'),
AerSimulator('aer_simulator_density_matrix'),
AerSimulator('aer_simulator_stabilizer'),
AerSimulator('aer_simulator_matrix_product_state'),
AerSimulator('aer_simulator_extended_stabilizer'),
AerSimulator('aer_simulator_unitary'), AerSimulator('aer_simulator_superop'),
QasmSimulator('qasm_simulator'), StatevectorSimulator('statevector_simulator'),
UnitarySimulator('unitary_simulator')]
```

0.2 Unitary Simulator

[36]:
```
circuit = QuantumCircuit(2)
circuit.h(0)
circuit.cx(0, 1)
circuit.draw('mpl')
```

[36]:



[37]:
```
unitary_simulator = Aer.get_backend('unitary_simulator')
unitary_simulator_result = unitary_simulator.run(circuit).result()
unitary_simulator_result.data()
```

[37]:
```
{'unitary': Operator([[ 0.70710678+0.00000000e+00j,  0.70710678-8.65956056e-17j,
                0.        +0.00000000e+00j,  0.        +0.00000000e+00j],
              [ 0.        +0.00000000e+00j,  0.        +0.00000000e+00j,
                0.70710678+0.00000000e+00j, -0.70710678+8.65956056e-17j],
              [ 0.        +0.00000000e+00j,  0.        +0.00000000e+00j,
                0.70710678+0.00000000e+00j,  0.70710678-8.65956056e-17j],
              [ 0.70710678+0.00000000e+00j, -0.70710678+8.65956056e-17j,
                0.        +0.00000000e+00j,  0.        +0.00000000e+00j]],
             input_dims=(2, 2), output_dims=(2, 2))}
```

[38]:
```
array_to_latex(unitary_simulator_result.get_unitary())
```

[38]:

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

0.3 Statevector Simulator

[27]:
```
circuit = QuantumCircuit(2)
circuit.h(0)
circuit.cx(0, 1)

statevector_simulator = Aer.get_backend('statevector_simulator')
statevector_simulator_result = statevector_simulator.run(circuit).result()
statevector_simulator_result.data()
```

[27]:
```
{'statevector': Statevector([0.70710678+0.j, 0.        +0.j, 0.        +0.j,
                 0.70710678+0.j],
                dims=(2, 2))}
```

[28]:
```
array_to_latex(statevector_simulator_result.get_statevector())
```

[28]:

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

[29]:
```
circuit = QuantumCircuit(2)
circuit.h(0)
circuit.save_statevector('breakpoint')
circuit.cx(0, 1)
```

```
statevector_simulator = Aer.get_backend('statevector_simulator')
statevector_simulator_result = statevector_simulator.run(circuit).result()
array_to_latex(statevector_simulator_result.data()['breakpoint'])
```

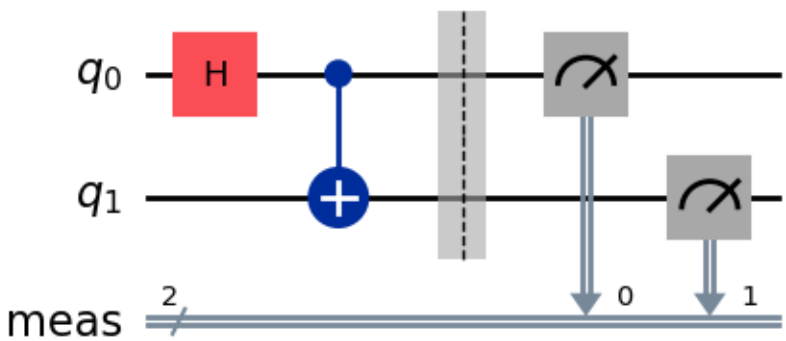
[29]:

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

0.4 QASM Simulator

[30]:
```
circuit = QuantumCircuit(2)
circuit.h(0)
circuit.cx(0, 1)
circuit.measure_all()  # <--
circuit.draw('mpl')
```

[30]:

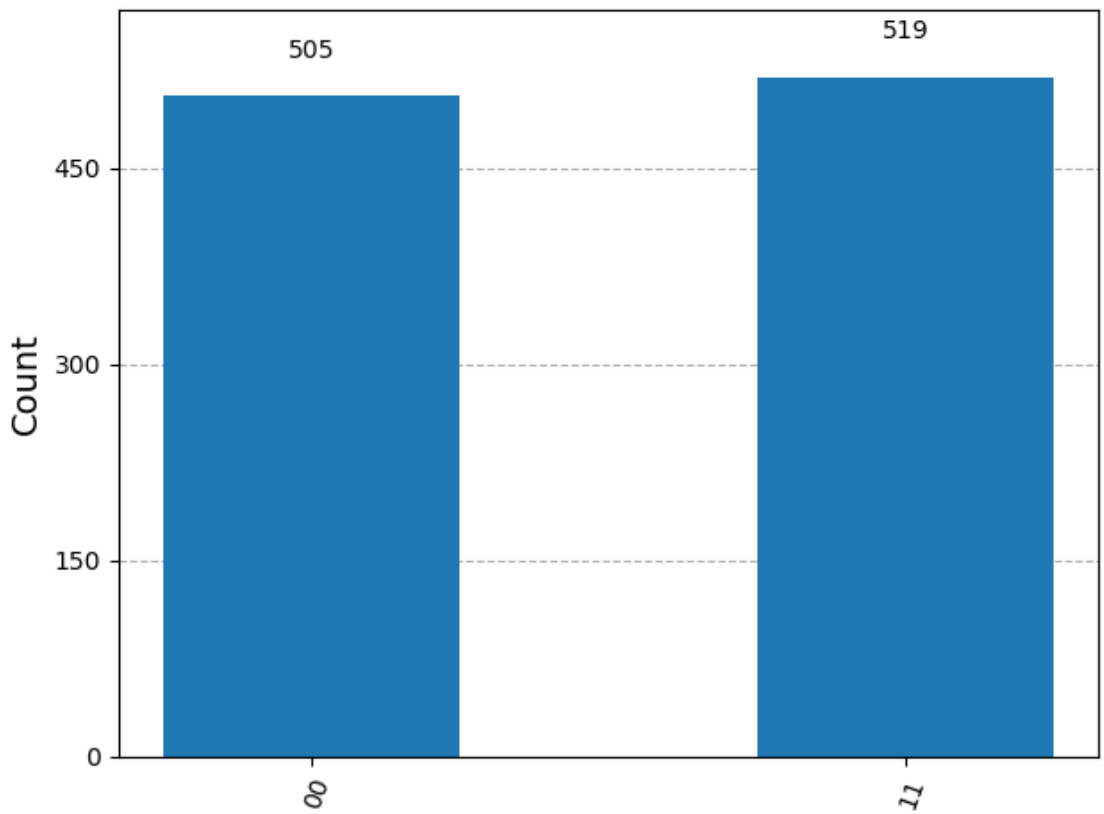


[31]:
```
qasm_simulator = Aer.get_backend('qasm_simulator')
qasm_simulator_result = qasm_simulator.run(circuit).result()
qasm_simulator_result.data()
```

[31]: {'counts': {'0x3': 519, '0x0': 505}}

[32]:
```
plot_histogram(qasm_simulator_result.get_counts())
```

[32]:



[33]:
```
memory_result = qasm_simulator.run(circuit, shots=10, memory=True).result()
memory_result.get_memory(circuit)
```

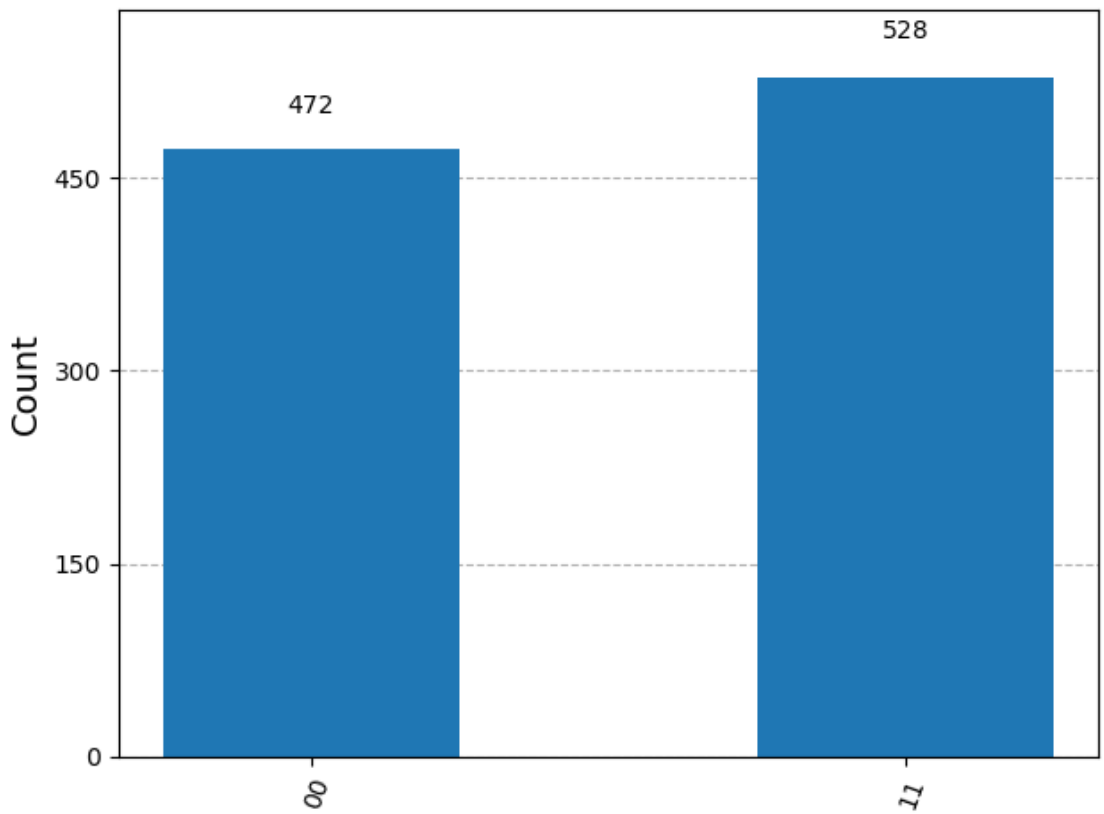
[33]: ['11', '00', '00', '00', '00', '11', '00', '00', '11', '00']

[47]:
```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()

backend = Aer.get_backend("aer_simulator")
tqc = transpile(qc, backend)
job = backend.run(tqc, shots=1000)
result = job.result()
counts = result.get_counts(tqc)
print(counts)
```

```
{'11': 528, '00': 472}
```

[48]:
```
plot_histogram(result.get_counts())
```

[48]:



[49]:
```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)

backend = Aer.get_backend("aer_simulator")
qc.save_statevector()
tqc = transpile(qc, backend)
job = backend.run(tqc)
result = job.result()
statevector = result.get_statevector(tqc, 4)
print(statevector)
```

```
Statevector([0.7071+0.j, 0.    +0.j, 0.    +0.j, 0.7071+0.j],
            dims=(2, 2))
```

[50]:
```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)

backend = Aer.get_backend("aer_simulator")
qc.save_unitary()
```

```
tqc = transpile(qc, backend)
job = backend.run(tqc)
result = job.result()
unitary = result.get_unitary(qc, 4)
print(unitary)
```

```
Operator([[ 0.7071+0.j,  0.7071-0.j,  0.    +0.j,  0.    +0.j],
          [ 0.    +0.j,  0.    +0.j,  0.7071+0.j, -0.7071+0.j],
          [ 0.    +0.j,  0.    +0.j,  0.7071+0.j,  0.7071-0.j],
          [ 0.7071+0.j, -0.7071+0.j,  0.    +0.j,  0.    +0.j]],
         input_dims=(2, 2), output_dims=(2, 2))
```

[51]:
```
array_to_latex(result.get_unitary())
```

[51]:

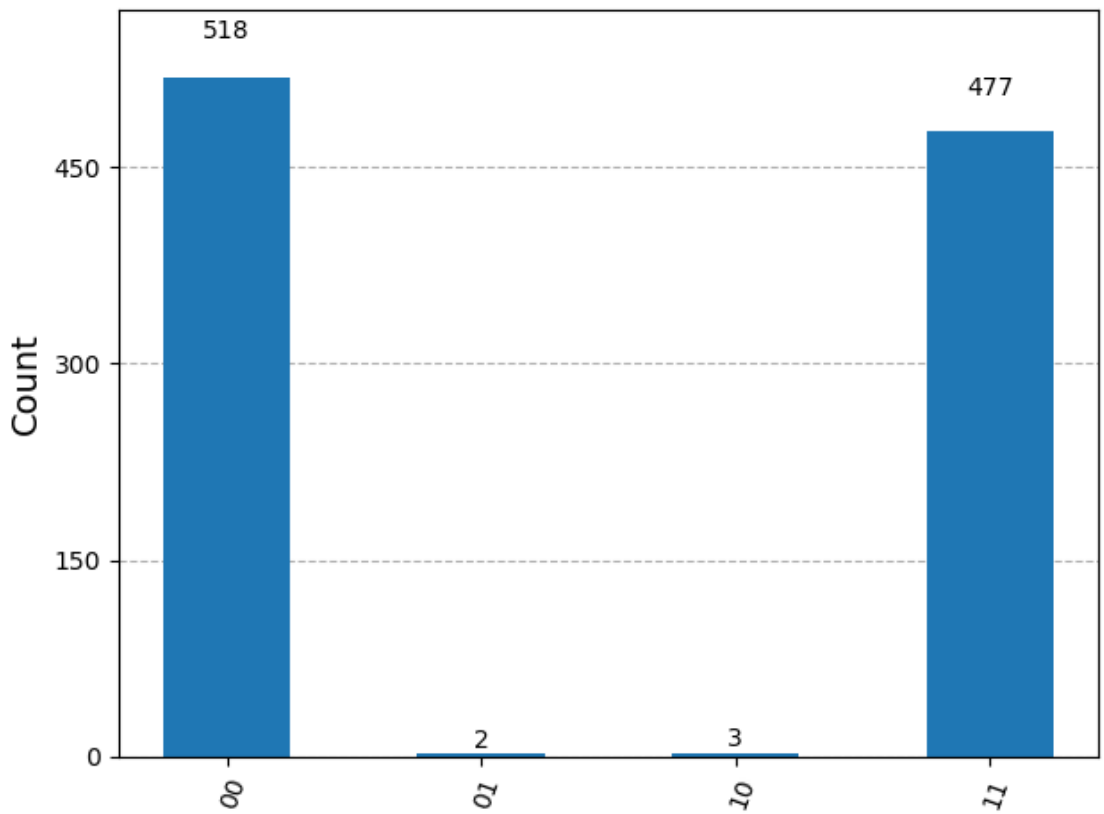
$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

[55]:
```
err_1 = depolarizing_error(0.95, 1)
err_2 = depolarizing_error(0.01, 2)
noise_model = NoiseModel()
noise_model.add_all_qubit_quantum_error(err_1,
                            ['u1', 'u2', 'u3'])
noise_model.add_all_qubit_quantum_error(err_2,
                                        ['cx'])

qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()

backend = Aer.get_backend("aer_simulator")
backend.set_options(noise_model=noise_model)
tqc = transpile(qc, backend)
job = backend.run(tqc, shots=1000)
result = job.result()
counts = result.get_counts(tqc)
print(counts)
```

```
{'01': 2, '11': 477, '10': 3, '00': 518}
```

[56]:
```
plot_histogram(result.get_counts())
```

[56]:

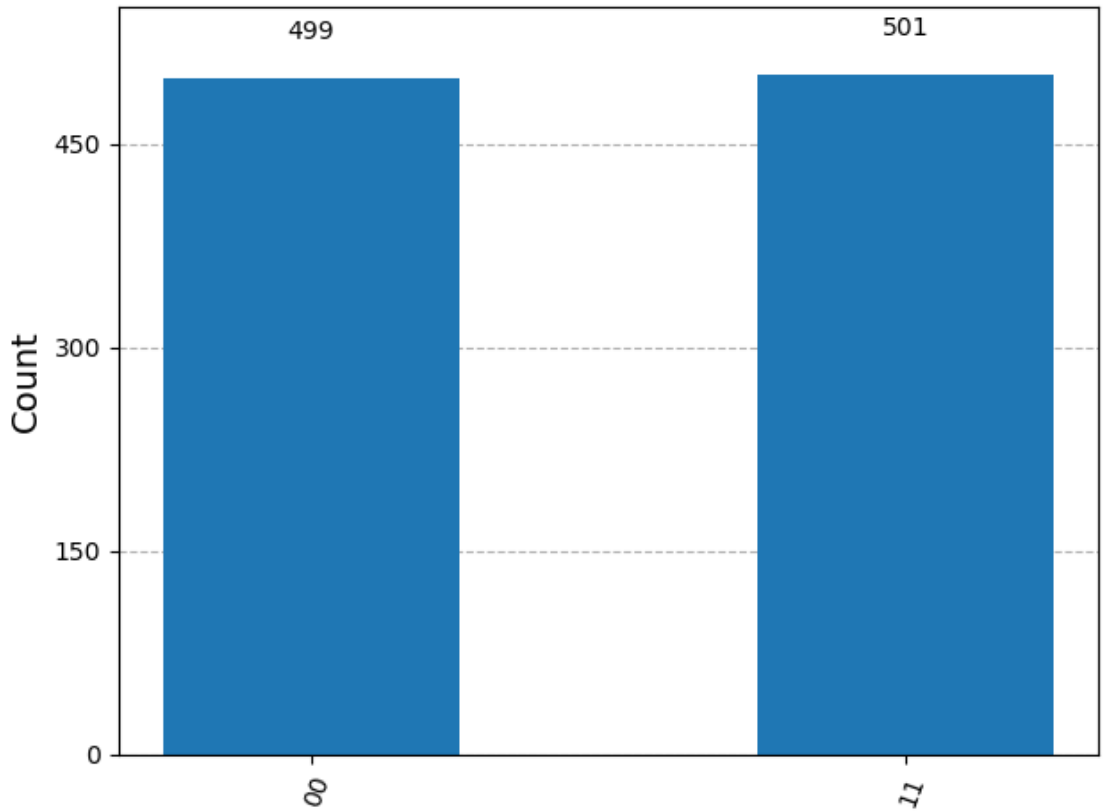


[58]:
```
# Your quantum circuit
qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0, 1)
qc.measure([0, 1], [0, 1])

# Initialize the simulator
simulator = AerSimulator()

# Transpile the circuit for the simulator
transpiled_qc = transpile(qc, simulator)

# Run the simulation
job = simulator.run(transpiled_qc, shots=1000)
result = job.result()
counts = result.get_counts(qc)
print(counts)
```

```
{'11': 501, '00': 499}
```

[59]:
```
plot_histogram(result.get_counts())
```

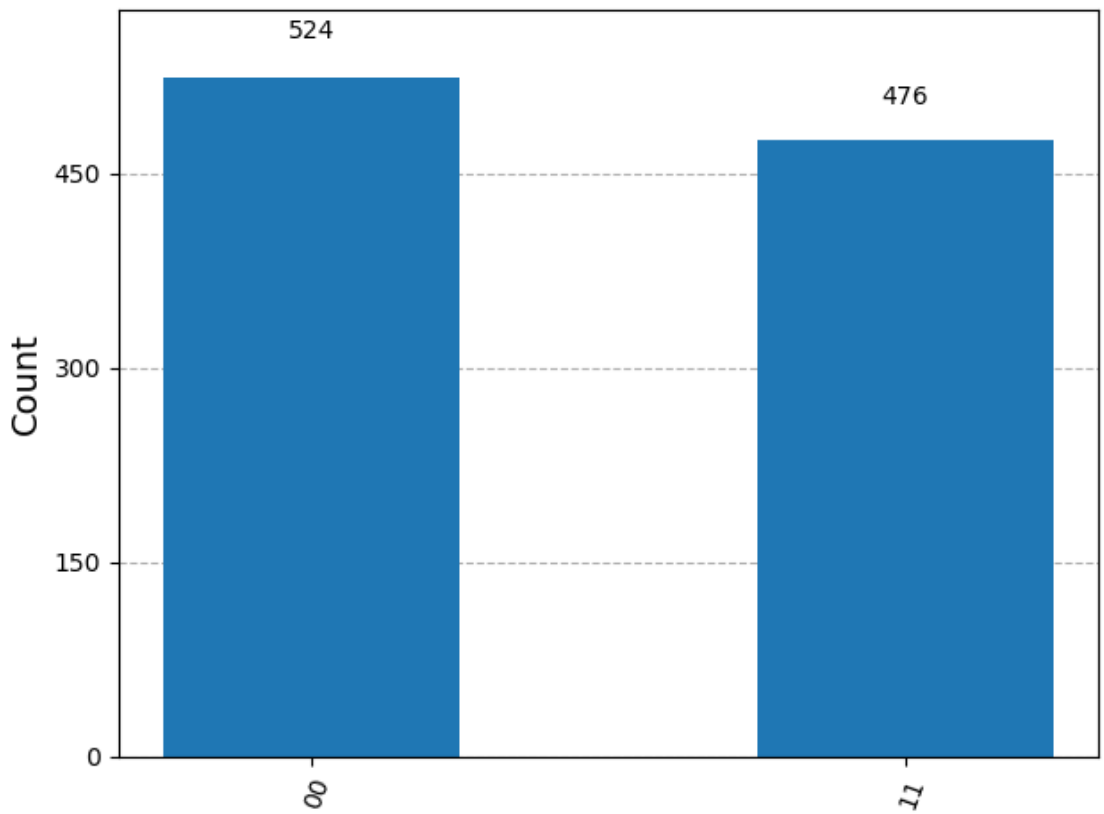
[59]:



[60]:

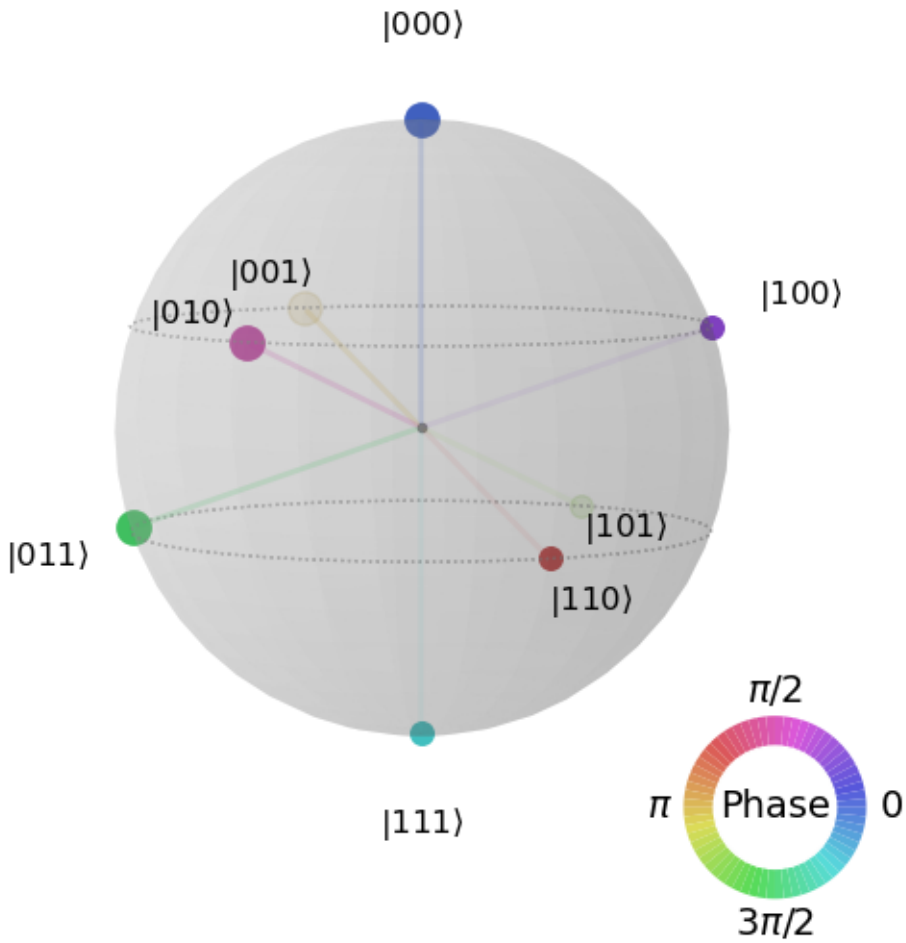
```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()

backend = Aer.get_backend("aer_simulator")
tqc = transpile(qc, backend)
job = backend.run(tqc, shots=1000)
result = job.result()
counts = result.get_counts(tqc)

plot_histogram(counts)
```

[60]:



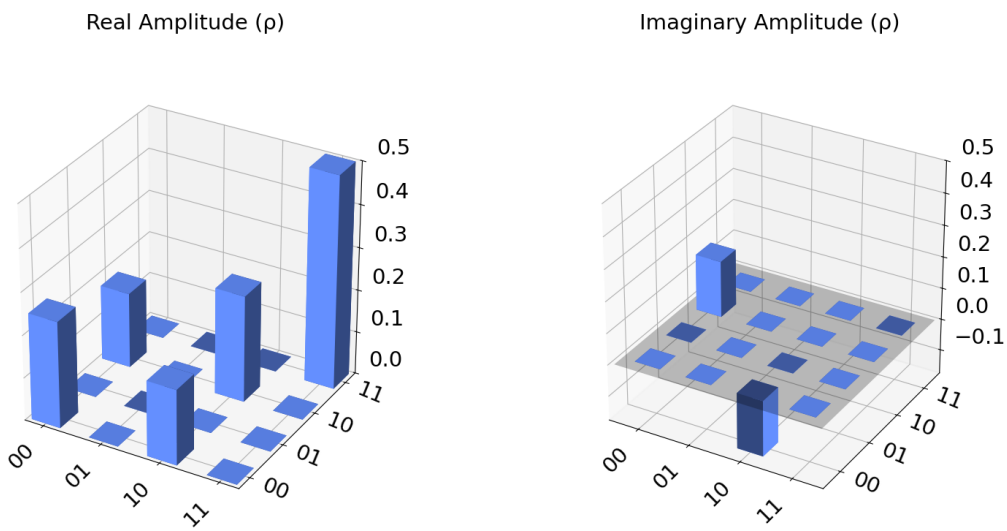
[63]:
```
backend = Aer.get_backend("aer_simulator_statevector")

qc = QuantumCircuit(3)
qc.rx(pi, 0)
qc.ry(pi/8, 2)
qc.swap(0, 2)
qc.h(0)
qc.cp(pi/2,0, 1)
qc.cp(pi/4, 0, 2)
qc.h(1)
qc.cp(pi/2, 1, 2)
qc.h(2)
qc.save_statevector()

tqc = transpile(qc, backend)
job = backend.run(tqc)
result = job.result()
statevector = result.get_statevector()

plot_state_qsphere(statevector)
```

[63]:



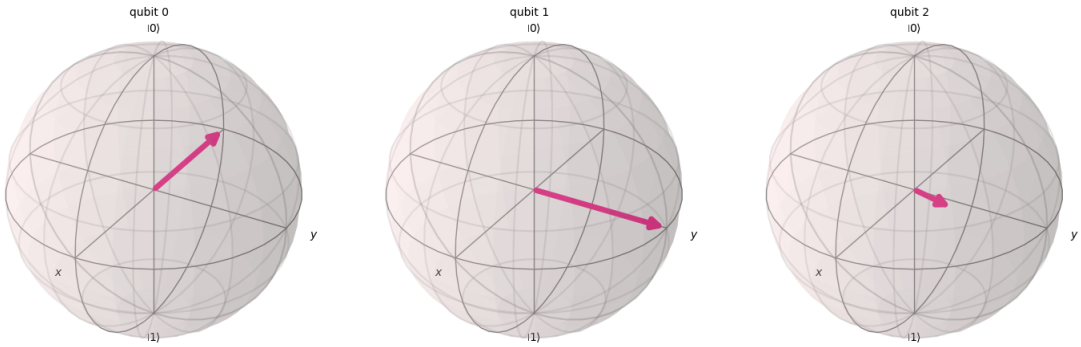
[66]:
```
dens_mat = 0.5*DensityMatrix.from_label('11') +  0.5*DensityMatrix.
 ↪from_label('+0')
tt_op = Operator.from_label('TT')
dens_mat = dens_mat.evolve(tt_op)
plot_state_city(dens_mat)
```

[66]:



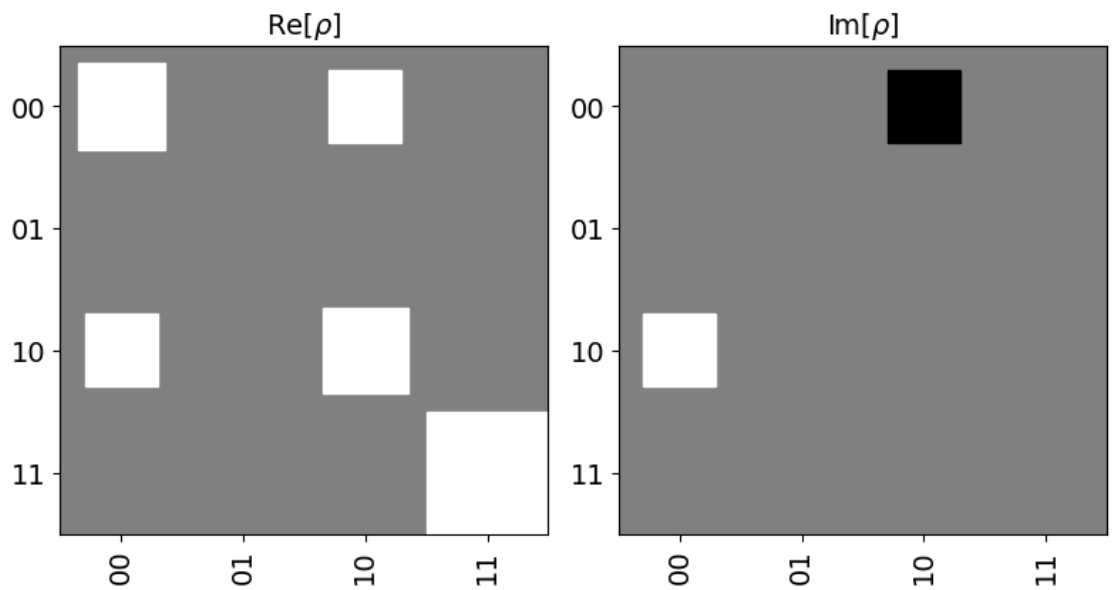
[69]:
```
backend = Aer.get_backend("aer_simulator_statevector")

qc = QuantumCircuit(3)
qc.rx(pi, 0)
qc.ry(pi/8, 2)
qc.swap(0, 2)
qc.h(0)
qc.cp(pi/2,0, 1)
qc.cp(pi/4, 0, 2)
qc.h(1)
qc.cp(pi/2, 1, 2)
qc.h(2)
qc.save_statevector()

tqc = transpile(qc, backend)
job = backend.run(tqc)
result = job.result()
statevector = result.get_statevector()

plot_bloch_multivector(statevector)
```

[69]:



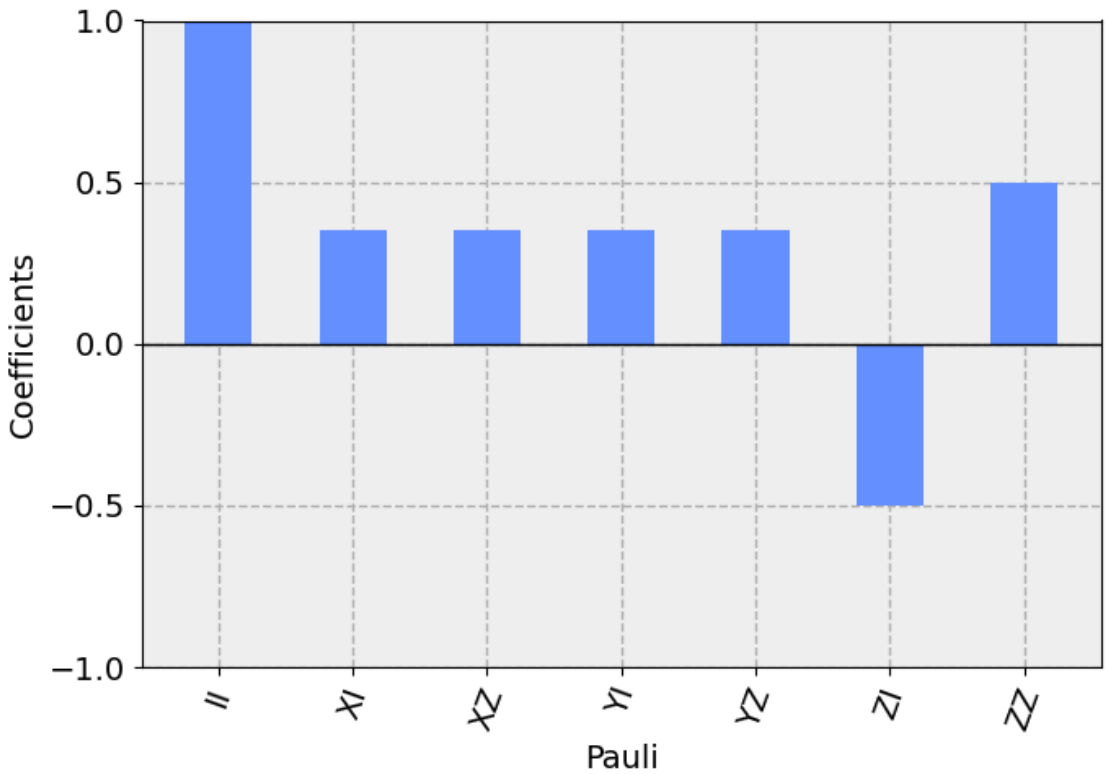
[72]:
```
dens_mat = 0.5*DensityMatrix.from_label('11') +  0.5*DensityMatrix.
 ↪from_label('+0')
tt_op = Operator.from_label('TT')
dens_mat = dens_mat.evolve(tt_op)

plot_state_hinton(dens_mat)
```

[72]:



[75]:
```
dens_mat = 0.5*DensityMatrix.from_label('11') +   0.5*DensityMatrix.
 ↪from_label('+0')
tt_op = Operator.from_label('TT')
dens_mat = dens_mat.evolve(tt_op)

plot_state_paulivec(dens_mat)
```

[75]:


1.0
0.5
0.0
−0.5
−1.0
Coefficients
II
XI
XZ
YI
YZ
ZI
ZZ
Pauli

```
[ ]:
```